Genuine Multipartite Entanglement Analysis

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Model
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Hubbard ring (1-D, PBC)

System parameters
-----------------
Sites (L)         : 4
Electrons (Ne)    : 4
Hopping (t)       : 10.0
Interaction (U)   : 1.0
Mode              : uniform
n-body order      : 4
```

```
Ground-state properties
-----------------------
E0                : -39.26003999
E1                : -39.25394536

<S2>              : 0.000000
S                 : 0.0000
<Sz>              : 0.000000
Multiplicity 2S+1 : 1.0
Spin state        : Singlet (S=0)

Double occupancy
----------------
<D> total         : 0.730035
<D> / site        : 0.182509
```

```
GME summary
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Extracted n-spin RDM : n = 4

Purity:
  gamma = Tr(rho^2)

Separable cut criterion:
  gamma_A = 1  <->  rho = rho_A x rho_B

C_GME (Troiani et al.):
  C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Decision logic:
  gamma=1, all Tr(rho_A^2)<1  -> GME
  gamma=1, some Tr(rho_A^2)=1 -> No GME
  gamma < 1                   -> Mixed

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Verification Test: n = N (pure state RDM)

When n = N, the nBRDM is a pure state.
The exact C_GME from eq.(1) and the lower bound from eqs.(6-9) should agree.

System

```
L  (total sites)   = 4
Ne (total electrons)= 4
n  (RDM body order) = 4
n = N ?             = YES
```

RDM properties

```
Tr(rho)    = 1.000000  (by construction)
Tr(rho^2)  = 1.0000000000
Pure state = YES ✓
```

Exact C_GME [eq.(1)]

```
C_GME (exact) = 0.9128709292
```

Basis-state lower bound [eqs.(6-9)]

```
I_Phi = 2*(sqrt(A) - sum_k sqrt(B_k))
C_LB  (basis) = 0.6666666667
Best z1 = [0, 1, 0, 1]
Best z2 = [1, 0, 1, 0]
Pairs checked = 240
```

Continuous optimization

```
Method : continuous_complex
C_LB   = 0.6666666667
Gain   = -0.00000000  vs basis
```

Comparison

```
C_exact         = 0.91287093
C_LB (basis)    = 0.66666667
C_LB (cont.)    = 0.66666667

Best diff = 2.46e-01

C_LB <= C_exact ? = YES ✓

Note: C_LB is a lower bound.
GHZ: C_LB = C_exact (tight).
Hubbard: C_LB < C_exact (loose).
```

```
RESULT: VERIFIED ✓
Lower bound is valid.
```

LOWER BOUND VERIFIED ✓